CHAPTER C3
DEAD LOADS, SOIL LOADS, AND HYDROSTATIC PRESSURE

C3.1 DEAD LOADS

C3.1.2 Weights of Materials and Constructions. To establish uniform practice among designers, it is desirable to present a list of materials generally used in building construction, together with their proper weights. Many building codes prescribe the minimum weights for only a few building materials, and in other instances no guide whatsoever is furnished on this subject. In some cases, the codes are drawn up so as to leave the question of what weights to use to the discretion of the building official, without providing any authoritative guide. This practice, as well as the use of incomplete lists, has been subjected to much criticism. The solution chosen has been to present, in this commentary, an extended list that will be useful to designer and official alike. However, special cases will unavoidably arise, and authority is therefore granted in the standard for the building official to deal with them.

For ease of computation, most values are given in terms of pounds per square foot, psf (kilonewtons per square meter, kN/m^2) of given thickness in Table C3.1-1a (Table C3.1-1b). Pounds-per-cubic-foot, lb/ft^3 (kN/m^3), values, consistent with the pounds-per-square foot psf (kilonewtons per square meter, kN/m^2) values, are also presented in some cases in Table C3.1-2. Some constructions for which a single figure is given actually have a considerable range in weight. The average figure given is suitable for general use, but when there is reason to suspect a considerable deviation from this, the actual weight should be determined.

Engineers, architects, and building owners are advised to consider factors that result in differences between actual and calculated loads.

Engineers and architects cannot be responsible for circumstances beyond their control. Experience has shown, however, that conditions are encountered that, if not considered in design, may reduce the future utility of a building or reduce its margin of safety. Among them are

1. Dead loads. There have been numerous instances in which the actual weights of members and construction materials have exceeded the values used in design. Care is advised in the use of tabular values. Also, allowances should be made for such factors as the influence of formwork and support deflections on the actual thickness of a concrete slab of prescribed nominal thickness.
2. Future installations. Allowance should be made for the weight of future wearing or protective surfaces where there is a good possibility that such may be applied. Special consideration should be given to the likely types and position of partitions, as insufficient provision for partitioning may reduce the future utility of the building.

Attention is directed also to the possibility of temporary changes in the use of a building, as in the case of clearing a dormitory for a dance or other recreational purpose.

C3.1.3 Weight of Fixed Service Equipment. Fixed service equipment includes but is not limited to plumbing stacks and risers; electrical feeders; heating, ventilating, and air conditioning systems; and process equipment such as vessels, tanks, piping, and cable trays. Both the empty weight of equipment and the maximum weight of contents are treated as dead load.

Section 1.3.6 indicates that when resistance to overturning, sliding, and uplift forces is provided by dead load, the dead load shall be taken as the minimum dead load likely to be in place. Therefore, liquid contents and movable trays shall not be used to resist these forces unless they are the source of the forces. For example, the liquid in a tank contributes to the seismic mass of the tank and therefore can be used to resist the seismic uplift; however, the weight of the liquid cannot be used to resist overturning, sliding, or uplift from wind loads because the liquid may not be present during the wind event.

C3.1.4 Vegetative and Landscaped Roofs. Landscaping elements, such as soil, plants, and drainage layer materials, and hardscaping elements, such as walkways, fences, and walls, are intended to remain in place and are therefore considered dead loads. While the weight of hardscaping materials does not fluctuate, the weight of soil and drainage layer materials used to support vegetative growth is subject to significant variation because of its ability to absorb and retain water. Where the weight is additive to other loads, the dead load should be computed assuming full saturation of soil and drainage layer materials. Where the weight acts to counteract uplift forces, the dead load should be computed assuming dry unit weight of the soil and dry drainage layer materials.

Vegetative and landscaped roof areas may be able to retain more water than the condition where the soil and drainage layer materials are fully saturated. The water may result from precipitation or from irrigation of the vegetation. This additional amount of water should be considered rain load in accordance with Chapter 8 or snow load in accordance with Chapter 7 as applicable.

C3.1.5 Solar Panels. This section clarifies that solar panel-related loads, including ballasted systems that are not permanently attached, shall be considered as dead loads for all load combinations specified in Chapter 2.

C3.2 SOIL LOADS AND HYDROSTATIC PRESSURE

C3.2.1 Lateral Pressures. Table 3.2-1 includes high earth pressures, 85 pcf (13.36 kN/m^2) or more, to show that certain soils are poor backfill material. In addition, when walls are

Table C3.1-1a Minimum Design Dead Loads (psf)[a]

Component	Load (psf)
CEILINGS	
Acoustical fiberboard	1
Gypsum board (per 1/8-in. thickness)	0.55
Mechanical duct allowance	4
Plaster on tile or concrete	5
Plaster on wood lath	8
Suspended steel channel system	2
Suspended metal lath and cement plaster	15
Suspended metal lath and gypsum plaster	10
Wood furring suspension system	2.5
COVERINGS, ROOF, AND WALL	
Asbestos-cement shingles	4
Asphalt shingles	2
Cement tile	16
Clay tile (for mortar add 10 psf)	
Book tile, 2-in.	12
Book tile, 3-in.	20
Ludowici	10
Roman	12
Spanish	19
Composition:	
Three-ply ready roofing	1
Four-ply felt and gravel	5.5
Five-ply felt and gravel	6
Copper or tin	1
Corrugated asbestos-cement roofing	4
Deck, metal, 20 gauge	2.5
Deck, metal, 18 gauge	3
Decking, 2-in. wood (Douglas fir)	5
Decking, 3-in. wood (Douglas fir)	8
Fiberboard, 1/2-in.	0.75
Gypsum sheathing, 1/2-in.	2
Insulation, roof boards (per inch thickness)	
Cellular glass	0.7
Fibrous glass	1.1
Fiberboard	1.5
Perlite	0.8
Polystyrene foam	0.2
Urethane foam with skin	0.5
Plywood (per 1/8-in. thickness)	0.4
Rigid insulation, 1/2-in.	0.75
Skylight, metal frame, 3/8-in. wire glass	8
Slate, 3/16-in.	7
Slate, 1/4-in.	10
Waterproofing membranes:	
Bituminous, gravel-covered	5.5
Bituminous, smooth surface	1.5
Liquid applied	1
Single-ply, sheet	0.7
Wood sheathing (per inch thickness)	3
Wood shingles	3
FLOOR FILL	
Cinder concrete, per inch	9
Lightweight concrete, per inch	8
Sand, per inch	8
Stone concrete, per inch	12
FLOORS AND FLOOR FINISHES	
Asphalt block (2-in.), 1/2-in. mortar	30
Cement finish (1-in.) on stone–concrete fill	32
Ceramic or quarry tile (3/4-in.) on 1/2-in. mortar bed	16
Ceramic or quarry tile (3/4-in.) on 1-in. mortar bed	23
Concrete fill finish (per inch thickness)	12
Hardwood flooring, 7/8-in.	4
Linoleum or asphalt tile, 1/4-in.	1
Marble and mortar on stone–concrete fill	33

continues

Table C3.1-1a (*Continued*)

Component					Load (psf)
Slate (per mm thickness)					15
Solid flat tile on 1-in. mortar base					23
Subflooring, 3/4-in.					3
Terrazzo (1-1/2-in.) directly on slab					19
Terrazzo (1-in.) on stone–concrete fill					32
Terrazzo (1-in.), 2-in. stone concrete					32
Wood block (3-in.) on mastic, no fill					10
Wood block (3-in.) on 1/2-in. mortar base					16
FLOORS, WOOD-JOIST (NO PLASTER)					
DOUBLE WOOD FLOOR					
Joint sizes (in.)	12-in. spacing (psf)	16-in. spacing (psf)	24-in. spacing (psf)		
2 × 6	6	5	5		
2 × 8	6	6	5		
2 × 10	7	6	6		
2 × 12	8	7	6		
FRAME PARTITIONS					
Movable steel partitions					4
Wood or steel studs, 1/2-in. gypsum board each side					8
Wood studs, 2 × 4, unplastered					4
Wood studs, 2 × 4, plastered one side					12
Wood studs, 2 × 4, plastered two sides					20
FRAME WALLS					
Exterior stud walls:					
2 × 4 @ 16-in., 5/8-in. gypsum, insulated, 3/8-in. siding					11
2 × 6 @ 16-in., 5/8-in. gypsum, insulated, 3/8-in. siding					12
Exterior stud walls with brick veneer					48
Windows, glass, frame, and sash					8
Clay brick wythes:					
4 in.					39
8 in.					79
12 in.					115
16 in.					155
Hollow concrete masonry unit wythes:					
Wythe thickness (in inches)	4	6	8	10	12
Density of unit (105 pcf) with grout spacing as follows:					
No grout	22	24	31	37	43
48 in. o.c.		29	38	47	55
40 in. o.c.		30	40	49	57
32 in. o.c.		32	42	52	61
24 in. o.c.		34	46	57	67
16 in. o.c.		40	53	66	79
Full grout		55	75	95	115
Density of unit (125 pcf) with grout spacing as follows:					
No grout	26	28	36	44	50
48 in. o.c.		33	44	54	62
40 in. o.c.		34	45	56	65
32 in. o.c.		36	47	58	68
24 in. o.c.		39	51	63	75
16 in. o.c.		44	59	73	87
Full grout		59	81	102	123
Density of unit (135 pcf) with grout spacing as follows:					
No grout	29	30	39	47	54
48 in. o.c.		36	47	57	66
40 in. o.c.		37	48	59	69
32 in. o.c.		38	50	62	72
24 in. o.c.		41	54	67	78
16 in. o.c.		46	61	76	90
Full grout		62	83	105	127
Solid concrete masonry unit wythes (incl. concrete brick):					
Wythe thickness (in mm)	4	6	8	10	12
Density of unit (105 pcf)	32	51	69	87	105
Density of unit (125 pcf)	38	60	81	102	124
Density of unit (135 pcf)	41	64	87	110	133

[a]Weights of masonry include mortar but not plaster. For plaster, add 5 psf for each face plastered. Values given represent averages. In some cases, there is a considerable range of weight for the same construction.

Table C3.1-1b Minimum Design Dead Loads (kN/m^2)[a]

Component	Load (kN/m^2)
CEILINGS	
Acoustical fiberboard	0.05
Gypsum board (per mm thickness)	0.008
Mechanical duct allowance	0.19
Plaster on tile or concrete	0.24
Plaster on wood lath	0.38
Suspended steel channel system	0.10
Suspended metal lath and cement plaster	0.72
Suspended metal lath and gypsum plaster	0.48
Wood furring suspension system	0.12
COVERINGS, ROOF, AND WALL	
Asbestos-cement shingles	0.19
Asphalt shingles	0.10
Cement tile	0.77
Clay tile (for mortar add 0.48 kN/m^2)	
Book tile, 51 mm	0.57
Book tile, 76 mm	0.96
Ludowici	0.48
Roman	0.57
Spanish	0.91
Composition:	
Three-ply ready roofing	0.05
Four-ply felt and gravel	0.26
Five-ply felt and gravel	0.29
Copper or tin	0.05
Corrugated asbestos-cement roofing	0.19
Deck, metal, 20 gauge	0.12
Deck, metal, 18 gauge	0.14
Decking, 51-mm wood (Douglas fir)	0.24
Decking, 76-mm wood (Douglas fir)	0.38
Fiberboard, 13 mm	0.04
Gypsum sheathing, 13 mm	0.10
Insulation, roof boards (per mm thickness)	
Cellular glass	0.0013
Fibrous glass	0.0021
Fiberboard	0.0028
Perlite	0.0015
Polystyrene foam	0.0004
Urethane foam with skin	0.0009
Plywood (per mm thickness)	0.006
Rigid insulation, 13 mm	0.04
Skylight, metal frame, 10-mm wire glass	0.38
Slate, 5 mm	0.34
Slate, 6 mm	0.48
Waterproofing membranes:	
Bituminous, gravel-covered	0.26
Bituminous, smooth surface	0.07
Liquid applied	0.05
Single-ply, sheet	0.03
Wood sheathing (per mm thickness)	
Plywood	0.0057
Oriented strand board	0.0062
Wood shingles	0.14
FLOOR FILL	
Cinder concrete, per mm	0.017
Lightweight concrete, per mm	0.015
Sand, per mm	0.015
Stone concrete, per mm	0.023
FLOORS AND FLOOR FINISHES	
Asphalt block (51 mm), 13-mm mortar	1.44
Cement finish (25 mm) on stone–concrete fill	1.53
Ceramic or quarry tile (19 mm) on 13-mm mortar bed	0.77
Ceramic or quarry tile (19 mm) on 25-mm mortar bed	1.10

continues

Table C3.1-1b (*Continued*)

Component					Load (kN/m^2)
Concrete fill finish (per mm thickness)					0.023
Hardwood flooring, 22 mm					0.19
Linoleum or asphalt tile, 6 mm					0.05
Marble and mortar on stone–concrete fill					1.58
Slate (per mm thickness)					0.028
Solid flat tile on 25-mm mortar base					1.10
Subflooring, 19 mm					0.14
Terrazzo (38 mm) directly on slab					0.91
Terrazzo (25 mm) on stone–concrete fill					1.53
Terrazzo (25 mm), 51-mm stone concrete					1.53
Wood block (76 mm) on mastic, no fill					0.48
Wood block (76 mm) on 13-mm mortar base					0.77
FLOORS, WOOD-JOIST (NO PLASTER)					
DOUBLE WOOD FLOOR					
Joint sizes (mm):	305-mm spacing (kN/m^2)	406-mm spacing (kN/m^2)	610-mm spacing (kN/m^2)		
51×152	0.29	0.24	0.24		
51×203	0.29	0.29	0.24		
51×254	0.34	0.29	0.29		
51×305	0.38	0.34	0.29		
FRAME PARTITIONS					
Movable steel partitions					0.19
Wood or steel studs, 13-mm gypsum board each side					0.38
Wood studs, 51×102, unplastered					0.19
Wood studs, 51×102, plastered one side					0.57
Wood studs, 51×102, plastered two sides					0.96
FRAME WALLS					
Exterior stud walls:					
51 mm $\times$ 102 mm@406 mm, 16-mm gypsum, insulated, 10-mm siding					0.53
51 mm $\times$ 152 mm@406 mm, 16-mm gypsum, insulated, 10-mm siding					0.57
Exterior stud walls with brick veneer					2.30
Windows, glass, frame, and sash					0.38
Clay brick wythes:					
102 mm					1.87
203 mm					3.78
305 mm					5.51
406 mm					7.42
Hollow concrete masonry unit wythes:					
Wythe thickness (in mm)	102	152	203	254	305
Density of unit (16.49 kN/m^3) with grout spacing as follows:					
No grout	1.05	1.29	1.68	2.01	2.35
1,219 mm		1.48	1.92	2.35	2.78
1,016 mm		1.58	2.06	2.54	3.02
813 mm		1.63	2.15	2.68	3.16
610 mm		1.77	2.35	2.92	3.45
406 mm		2.01	2.68	3.35	4.02
Full grout		2.73	3.69	4.69	5.70
Density of unit (19.64 kN/m^3) with grout spacing as follows:					
No grout	1.25	1.34	1.72	2.11	2.39
1,219 mm		1.58	2.11	2.59	2.97
1,016 mm		1.63	2.15	2.68	3.11
813 mm		1.72	2.25	2.78	3.26
610 mm		1.87	2.44	3.02	3.59
406 mm		2.11	2.78	3.50	4.17
Full grout		2.82	3.88	4.88	5.89
Density of unit (21.21 kN/m^3) with grout spacing as follows:					
No grout	1.39	1.68	2.15	2.59	3.02
1,219 mm		1.70	2.39	2.92	3.45
1,016 mm		1.72	2.54	3.11	3.69
813 mm		1.82	2.63	3.26	3.83
610 mm		1.96	2.82	3.50	4.12
406 mm		2.25	3.16	3.93	4.69
Full grout		3.06	4.17	5.27	6.37

continues

Table C3.1-1b (*Continued*)

Component					Load (kN/m^2)
Solid concrete masonry unit					
Wythe thickness (in mm)	102	152	203	254	305
Density of unit ($16.49\ kN/m^3$)	1.53	2.35	3.21	4.02	4.88
Density of unit ($19.64\ kN/m^3$)	1.82	2.82	3.78	4.79	5.79
Density of unit ($21.21\ kN/m^3$)	1.96	3.02	4.12	5.17	6.27

[a]Weights of masonry include mortar but not plaster. For plaster, add $0.24\ kN/m^3$ for each face plastered. Values given represent averages. In some cases, there is a considerable range of weight for the same construction.

Table C3.1-2 Minimum Densities for Design Loads from Materials

Material	Density (lb/ft^3)	Density (kN/m^3)
Aluminum	170	27
Bituminous products		
Asphaltum	81	12.7
Graphite	135	21.2
Paraffin	56	8.8
Petroleum, crude	55	8.6
Petroleum, refined	50	7.9
Petroleum, benzine	46	7.2
Petroleum, gasoline	42	6.6
Pitch	69	10.8
Tar	75	11.8
Brass	526	82.6
Bronze	552	86.7
Cast-stone masonry (cement, stone, sand)	144	22.6
Cement, Portland, loose	90	14.1
Ceramic tile	150	23.6
Charcoal	12	1.9
Cinder fill	57	9.0
Cinders, dry, in bulk	45	7.1
Coal		
Anthracite, piled	52	8.2
Bituminous, piled	47	7.4
Lignite, piled	47	7.4
Peat, dry, piled	23	3.6
Concrete, plain		
Cinder	108	17.0
Expanded-slag aggregate	100	15.7
Haydite (burned-clay aggregate)	90	14.1
Slag	132	20.7
Stone (including gravel)	144	22.6
Vermiculite and perlite aggregate, nonload-bearing	25–50	3.9–7.9
Other light aggregate, load-bearing	70–105	11.0–16.5
Concrete, reinforced		
Cinder	111	17.4
Slag	138	21.7
Stone (including gravel)	150	23.6
Copper	556	87.3
Cork, compressed	14	2.2
Earth (not submerged)		
Clay, dry	63	9.9
Clay, damp	110	17.3
Clay and gravel, dry	100	15.7
Silt, moist, loose	78	12.3

Table C3.1-2 (*Continued*)

Material	Density (lb/ft^3)	Density (kN/m^3)
Silt, moist, packed	96	15.1
Silt, flowing	108	17.0
Sand and gravel, dry, loose	100	15.7
Sand and gravel, dry, packed	110	17.3
Sand and gravel, wet	120	18.9
Earth (submerged)		
Clay	80	12.6
Soil	70	11.0
River mud	90	14.1
Sand or gravel	60	9.4
Sand or gravel and clay	65	10.2
Glass	160	25.1
Gravel, dry	104	16.3
Gypsum, loose	70	11.0
Gypsum, wallboard	50	7.9
Ice	57	9.0
Iron		
Cast	450	70.7
Wrought	480	75.4
Lead	710	111.5
Lime		
Hydrated, loose	32	5.0
Hydrated, compacted	45	7.1
Masonry, ashlar stone		
Granite	165	25.9
Limestone, crystalline	165	25.9
Limestone, oolitic	135	21.2
Marble	173	27.2
Sandstone	144	22.6
Masonry, brick		
Hard (low absorption)	130	20.4
Medium (medium absorption)	115	18.1
Soft (high absorption)	100	15.7
Masonry, concrete[a]		
Lightweight units	105	16.5
Medium weight units	125	19.6
Normal weight units	135	21.2
Masonry grout	140	22.0
Masonry, rubble stone		
Granite	153	24.0
Limestone, crystalline	147	23.1
Limestone, oolitic	138	21.7
Marble	156	24.5
Sandstone	137	21.5

continues

Table C3.1-2 (*Continued*)

Material	Density (lb/ft³)	Density (kN/m³)
Mortar, cement or lime	130	20.4
Particleboard	45	7.1
Plywood	36	5.7
Riprap (not submerged)		
Limestone	83	13.0
Sandstone	90	14.1
Sand		
Clean and dry	90	14.1
River, dry	106	16.7
Slag		
Bank	70	11.0
Bank screenings	108	17.0
Machine	96	15.1
Sand	52	8.2
Slate	172	27.0
Steel, cold-drawn	492	77.3
Stone, quarried, piled		
Basalt, granite, gneiss	96	15.1
Limestone, marble, quartz	95	14.9
Sandstone	82	12.9
Shale	92	14.5
Greenstone, hornblende	107	16.8
Terra cotta, architectural		
Voids filled	120	18.9
Voids unfilled	72	11.3
Tin	459	72.1
Water		
Fresh	62	9.7
Sea	64	10.1
Wood, seasoned		
Ash, commercial white	41	6.4
Cypress, southern	34	5.3
Fir, Douglas, coast region	34	5.3
Hem fir	28	4.4
Oak, commercial reds and whites	47	7.4
Pine, southern yellow	37	5.8
Redwood	28	4.4
Spruce, red, white, and Sitka	29	4.5
Western hemlock	32	5.0
Zinc, rolled sheet	449	70.5

[a]Tabulated values apply to solid masonry and to the solid portion of hollow masonry.

unyielding, the earth pressure is increased from active pressure toward earth pressure at rest, resulting in 60 pcf (9.43 kN/m^3) for granular soils and 100 pcf (15.71 kN/m^3) for silt and clay type soils (Terzaghi and Peck 1967). Examples of light floor systems supported on shallow basement walls mentioned in Table 3.2-1 are floor systems with wood joists and flooring and cold-formed steel joists without a cast-in-place concrete floor attached.

Expansive soils exist in many regions of the United States and may cause serious damage to basement walls unless special design considerations are provided. Expansive soils should not be used as backfill because they can exert very high pressures against walls. Special soil testing is required to determine the magnitude of these pressures. It is preferable to excavate expansive soil and backfill with nonexpansive, freely draining sands or gravels. The excavated back slope adjacent to the wall should be no steeper than 45° from the horizontal to minimize the transmission of swelling pressure from the expansive soil through the new backfill. Other special details are recommended, such as a cap of nonpervious soil on top of the backfill and provision of foundation drains. Refer to current reference books on geotechnical engineering for guidance.

C3.2.2 Uplift Loads on Floors and Foundations. If expansive soils are present under floors or footings, large pressures can be exerted and must be resisted by special design. Alternatively, the expansive soil can be removed and replaced with nonexpansive material. A geotechnical engineer should make recommendations in these situations.

REFERENCE

Terzaghi, K., and Peck, R. B. (1967). *Soil mechanics in engineering practice*, 2nd Ed., John Wiley & Sons, New York.

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